

El conjunto de ceros de una familia de polinomios es igual al conjunto de ceros del ideal que genera

Lema 1. Sea K un cuerpo y $\mathcal{F} \subset K[x_1, \dots, x_n]$ no vacío, entonces $\mathbb{V}(\mathcal{F}) = \mathbb{V}(\langle \mathcal{F} \rangle)$.

Demostración: Veamos ambos contenidos:

$\supset$ Viene dada por la obs-con-ceros-ideal-generado-contenido-trivial/??.

$\subset$ Si $\mathbb{V}(\mathcal{F}) = \emptyset$, entonces $\mathbb{V}(\langle \mathcal{F} \rangle) = \emptyset$ y hemos terminado. Supongamos que $\mathbb{V}(\mathcal{F}) \neq \emptyset$. Sea $a = (a_1, \dots, a_n) \in \mathbb{V}(\mathcal{F})$, entonces $\forall p \in \mathcal{F} : p(a) = 0$. Sea $q \in \langle \mathcal{F} \rangle$, entonces

$$\exists p_1, \dots, p_r \in \mathcal{F}, s_1, \dots, s_r \in K[x_1, \dots, x_n] : q = \sum_{i=1}^r s_i p_i \quad (\text{lem-ideal-generado/??}).$$

Por tanto, $q(a) = \sum_{i=1}^r s_i(a) p_i(a) = \sum_{i=1}^r s_i(a) \cdot 0 = 0$, luego $a \in \mathbb{V}(\langle \mathcal{F} \rangle)$. ■

Referenciado en

- prop-variedad-algebraica-afin-ideal

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